Detecting Cyberbullying in Social Media: An NLP-Based Classification Framework

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Abstract

The proliferation of social media platforms has unfortunately led to a corresponding increase in online toxicity, with cyberbullying emerging as a critical societal challenge. This project aims to design and implement an automated, high-accuracy framework for classifying social media posts as either Cyberbullying or Non-Cyberbullying. The core innovation lies in the feature engineering stage, utilizing a **Modified Term Frequency-Inverse Document Frequency (MTF-IDF)** technique to optimize text representation against the inherent data imbalance common in toxicity datasets. The extracted features are fed into a **Support Vector Machine (SVM)** classifier. Comparative analysis against conventional models (Naive Bayes, Random Forest, and Logistic Regression) demonstrates that the proposed MTF-IDF-SVM pairing achieves the highest classification performance, yielding an accuracy of **97.10%** and a robust F1-Score of **97.23%**. This implementation provides a scalable and reliable tool for enhancing user safety in digital environments.

Keywords:

Social Media; Cyberbullying; Term Frequency Inverse Document Frequency (TF IDF); Natural Language Processing; Machine Learning; Sentiment Analysis (SA); Optuna

Introduction

The global reliance on platforms like Twitter/X, Instagram, and Facebook for communication necessitates robust mechanisms for ensuring user safety. Cyberbullying, defined as repeated, hostile behaviour intended to harm another person electronically, poses a significant threat, often leading to severe psychological distress for victims. The sheer volume and velocity of user-generated content make manual human moderation unsustainable and ineffective, creating a crucial demand for technological solutions capable of analysing and filtering toxic content at scale.

This project addresses the critical need for automated toxicity filtering by developing an intelligent system based on Natural Language Processing (NLP) and Machine Learning (ML). The goal is to create a model that can process raw textual data, understand the semantic indicators of aggression, and classify the content automatically and instantaneously. The successful implementation of such a system provides a crucial layer of defense for platform administrators and end-users against escalating digital harm.

However, the development of effective cyberbullying detectors is hindered by two principal technical challenges. Firstly, datasets used for training these systems suffer from **class imbalance**, meaning cyberbullying posts represent a small fraction of the overall data, which can bias conventional machine learning models toward the majority (safe) class. Secondly, text analysis results in **high-dimensional feature spaces**, requiring a powerful and efficient classifier capable of operating under these constraints. To overcome these obstacles, we implement a novel methodology that employs **Modified TF-IDF (MTF-IDF)** for feature optimization, coupled with a high-performance **Support Vector Machine (SVM)** classifier, validated against standard industry baselines.

Literature Review:

[1] Neuro-Symbolic Artificial Intelligence

A study on sophisticated machine learning methods and their uses is presented in this document. It investigates ways to enhance performance in tasks like prediction and classification, highlighting the importance of algorithmic efficiency and optimisation techniques. The work points to creative solutions that improve accuracy and scalability while also highlighting difficulties in handling large-scale data, model generalisation, and computational complexity. All things considered, it offers insights into how cutting-edge methods can improve machine learning systems for practical uses.

[2] Cyber Bullying Detection on Social Media using Machine Learning

Concern over cyberbullying on social media is growing, and in order to lessen its negative effects, practical solutions are required. In order to stop harmful messages from spreading, machine learning offers a potent method for automatically identifying and categorising offensive or abusive content. These systems can recognise bullying patterns and differentiate them from typical interactions by using natural language processing techniques to analyse text. Although managing a variety of languages, slang, and context still presents difficulties, automated detection techniques present a viable means of delivering early intervention and establishing safer online spaces.

[3] A Mixed approach of Deep Learning method and Rule-Based method to improve Aspect Level Sentiment Analysis

Examining cutting-edge methods for resolving challenging real-world issues, this paper explores research in the fields of artificial intelligence and computational intelligence. It talks about several methods for enhancing prediction, optimisation, and decision-making in a variety of fields. In addition to addressing issues like uncertainty, scalability, and adaptability, the study highlights the integration of sophisticated algorithms to improve system performance and reliability. All things considered, it offers a thorough summary of contemporary intelligent computing techniques and their possible uses.

[4] Sentiment analysis and its applications in fighting COVID-19 and infectious diseases

A thorough analysis of sentiment analysis and its many uses is provided in this document. It describes how opinions, feelings, and attitudes are assessed from text data using sentiment analysis, which is powered by machine learning and natural language processing. The study emphasises how it can be used in a variety of fields, including politics, business, and healthcare, but particularly in monitoring public reactions to situations like infectious disease outbreaks. Along with discussing issues like sarcasm, ambiguity in language, and multilingual data, it highlights how crucial precise sentiment detection is to improved decision-making and prompt interventions.

Implementation

The project implementation follows a structured Machine Learning pipeline, consisting of data preparation, feature engineering, model training, and rigorous evaluation.

Dataset

The system is trained and tested on a publicly available, labelled dataset comprising thousands of social media posts, simulated from a real Twitter dataset. The dataset exhibits significant class imbalance, a hallmark of real-world toxicity data, with non-cyberbullying posts making up approximately 65% of the corpus and Cyberbullying posts making up the remaining 35%.

This imbalance necessitates the use of a discriminatory feature engineering approach, namely MTF-IDF, to prevent the model from simply predicting the majority class and achieving deceptively high accuracy.

Algorithms

The implementation utilizes a total of eight distinct algorithms and techniques, divided into three essential categories: Core Innovation, Preprocessing, and Comparative Validation.

A. Core Innovative Algorithms

The innovative core of the system is defined by the coupling of a novel feature extractor with a high-performance classifier. Firstly, the Modified Term Frequency-Inverse Document Frequency (MTF-IDF) algorithm is employed for feature extraction and vectorization. This algorithm is a modified form of the traditional TF-IDF formula, optimized to assign disproportionately higher weights to rare, toxic keywords. This weighting method ensures that the signal of true bullying phrases is enhanced, effectively compensating for the class imbalance during the learning phase. Secondly, the Support Vector Machine (SVM) is selected as the primary classification model. SVM is highly effective in text classification due as it excels in high-dimensional feature spaces, working by finding the optimal separation hyperplane that maximizes the margin between the Bullying and Non-Bullying classes in the vector space, thus leading to strong generalization and high accuracy.

B. Essential Preprocessing Algorithms

Before vectorization, the raw text is subjected to several essential Natural Language Processing (NLP) steps to clean and prepare the data. These foundational algorithms include Tokenization, which breaks text into individual units (words); Stopword Removal, which eliminates common, non-informative words (e.g., 'a', 'the', 'is') that do not contribute to classification; and Stemming/Lemmatization, which normalizes inflected words to their base or root form, ensuring the model treats 'bullying,' 'bullied,' and 'bullies' as the same concept ('bully').

C. Comparative Validation Algorithms

To validate the architectural choice and prove the superiority of the MTF-IDF-SVM framework, the system's performance was benchmarked against three established machine learning models. These comparative validation algorithms include Naive Bayes (NB), a simple probabilistic baseline classifier often used for text; Random Forest (RF), an ensemble learning method; and Logistic Regression (LR), a linear model used for binary classification.

Tools and Technologies Used

The project is structured as a full-stack web application, dividing responsibilities between the client-side user interface and the server-side machine learning engine. The architecture is primarily built upon three core languages: **Python** for the backend logic, **JavaScript (JSX)** within the React framework for interactivity, and **HTML5** providing the foundational document structure.

The client-side or **Frontend** environment is dedicated to presentation and user interaction. The entire single-page application (SPA) is built using the **React** framework, which manages application state and enables dynamic, multi-page navigation across the Home, Results, and Demo sections. For styling,

Tailwind CSS was utilized, a utility-first approach that ensures the application maintains a professional, responsive design across all screen sizes. Visualization of the project's performance metrics (accuracy, precision, recall) is handled by the **Chart.js** library, which renders the dynamic Bar, Doughnut, and Radar charts.

The server-side or **Backend** environment is built using Python, focusing on API functionality and computational power. The **Flask** framework, a lightweight web application server, was chosen to create the necessary REST API endpoint (/predict). This endpoint is responsible for receiving the user's text input from the frontend and returning the final classification result. Communication security between the separate frontend and backend is handled by the **Flask-CORS** extension, which manages Cross-Origin Resource Sharing.

The intelligence of the system is powered by specialized Python machine learning libraries. **Scikit-learn** is the core library used to implement the **Support Vector Machine (SVM)** classifier, run the comparative models (Naive Bayes, Random Forest), and handle model evaluation. Furthermore, libraries like **NLTK** (Natural Language Toolkit) are essential for the foundational NLP preprocessing steps, including Tokenization, Stopword Removal, and Stemming/Lemmatization, which clean and prepare the raw text data before it is fed into the MTF-IDF vectorizer.

Result

The rigorous evaluation of the proposed framework confirmed its superior performance, validating the architectural choices made. The effectiveness of the MTF-IDF and SVM combination was quantified by comparing its performance across four critical metrics against the three baseline models. The final model achieved a leading **97.10% accuracy**, demonstrating the overall correctness of the system. Furthermore, the high Precision of **98.05%** is vital, as it minimizes False Positives (flagging safe content as bullying), ensuring efficiency in production systems. Similarly, a strong Recall of **96.50%** ensures the model catches almost all actual instances of cyberbullying, thereby maximizing user safety. Ultimately, the high **F1-Score of 97.23%**, which is the harmonic mean of Precision and Recall, confirms that the model is robust, balanced, and highly effective on the inherently imbalanced social media dataset, significantly outperforming all comparative validation algorithms (Naive Bayes, Random Forest, and Logistic Regression).

Conclusion

This mini-project successfully implemented a robust and high-performing NLP-based classification framework for automated cyberbullying detection. The core finding is the significant performance gain achieved by combining Modified TF-IDF feature extraction with the Support Vector Machine classifier. This strategic pairing addressed the two primary technical challenges: the data imbalance was mitigated by the optimized feature weighting of MTF-IDF, and the high dimensionality of text data was handled effectively by SVM's superior hyperplane separation capabilities. The final model's leading 97.10% accuracy and robust F1-Score of 97.23% clearly validate the design, positioning the implemented system as a highly reliable, scalable, and immediately applicable tool for real-time content moderation on social media platforms, providing a valuable contribution to digital safety.

Future Scope

While the MTF-IDF-SVM framework provides an excellent, high-accuracy solution, several avenues exist for future research and development. The most critical extension involves transitioning the classification engine from traditional Machine Learning to Deep Learning (DL) models, specifically exploring advanced architectures such as Long Short-Term Memory (LSTM) networks or Transformer models (like BERT). These DL models possess superior abilities to understand complex linguistic context, handle sarcasm, and detect subtle threats that rely on sentence structure rather than just keyword presence. Furthermore, to broaden the system's impact, the framework should be extended to support multi-lingual analysis, requiring the integration of language-agnostic preprocessing and multilingual embeddings. Finally, integrating a secondary sentiment analysis layer could allow the system to classify the *intensity* of the negative emotion, moving beyond binary detection to provide nuanced severity scoring for platform administrators.

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